

Project Design Phase

Problem – Solution Fit Template

Date	5 September 2026
Team ID	SWUID2026191970
Project Name	Metro Ticket Generating System in ServiceNow

Problem – Solution Fit

The Problem–Solution Fit identifies the main commuter problems in metro ticket booking and shows how the proposed ServiceNow solution directly addresses them.

Purpose

- Reduce waiting time and manual effort during metro ticket booking.
- Provide a simple digital booking experience through ServiceNow.
- Automate fare calculation and reduce calculation errors.
- Generate QR-based digital tickets for convenient access.
- Support digital payment options such as UPI and cards.
- Reduce paper usage and improve operational efficiency.

Metro Ticket Generating System – Problem–Solution Fit Canvas

CS CUSTOMER SEGMENTS Metro commuters, daily office/college travelers, occasional passengers and users who need quick point-to-point tickets.	CC CUSTOMER CONSTRAINTS Long queues, manual ticket purchase, fare calculation difficulty, physical ticket handling and limited convenience during busy hours.	AS AVAILABLE SOLUTION ServiceNow ticket booking with source, destination, journey type, passenger count, automated fare calculation and QR ticket generation.
JP JOB-TO-BE-DONE / PROBLEMS Book a metro ticket quickly, select the correct route, know the fare, choose payment and receive a ticket without unnecessary waiting.	RC PROBLEM ROOT CAUSE Traditional/manual ticketing creates queues and repeated human effort. Manual fare handling may cause delays or calculation mistakes.	BH BEHAVIORS Passengers prefer fast self-service booking, clear station selection, immediate fare information, digital payment and easy mobile ticket access.
TR TRIGGERS Peak-hour travel, urgent journeys, long queues, lack of cash, return journeys or the need to reach the station gate quickly.	SL YOUR SOLUTION Users select Starting From and Going To, choose Single/Return Journey and passenger count, view fare, select payment mode and receive a QR-based digital ticket.	CH CHANNELS OF AWARENESS ServiceNow portal/catalog interface, metro station displays, QR awareness posters, mobile access and metro support channels.
BE CUSTOMER EXPERIENCE BEFORE Manual/physical ticketing, waiting in queues, fare enquiry and ticket handling.	AE CUSTOMER EXPERIENCE AFTER Digital booking, automated fare calculation, digital payment selection and QR ticket access.	OP AREAS OF OPPORTUNITY Real station/fare data, QR validation, notifications and future real-time metro service integration.

References

1. <https://www.ideahackers.network/problem-solution-fit-canvas/>
2. <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>